

MUST lecture 8

Principal Components Analysis, PCA **(Dimension reduction by decorrelating variables)**

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Principal Components Analysis

Motivation Given random vector $X = [X_1, X_2, \dots, X_p]^T$, $Cov[X] = \Sigma$

Often (some of) the variables are correlated and thus to some degree giving the same information

Idea Define **new uncorrelated variables (Principal Components, PC)** as linear combinations of the original variables such that the variance (information) of the PC's are maximized

Formally $PC_1 \stackrel{\text{def}}{=} a_{11}X_1 + \dots + a_{1p}X_p = a_1^T X$ under the constraints
 a_1 coefficients are chosen to maximize $V[PC_1] = a_1^T \Sigma a_1$
 $\|a_1\| = 1$ (unit vector)

$PC_2 \stackrel{\text{def}}{=} a_{21}X_1 + \dots + a_{2p}X_p = a_2^T X$ under the constraints
 a_2 coefficients are chosen to maximize $V[PC_2] = a_2^T \Sigma a_2$
 $\|a_2\| = 1$ (unit vector)
 $Cov[PC_1, PC_2] = 0$ (key point)

$\vdots$

$PC_p \stackrel{\text{def}}{=} a_{p1}X_1 + \dots + a_{pp}X_p = a_p^T X$ under the constraints
 a_p coefficients are chosen to maximize $V[PC_p] = a_p^T \Sigma a_p$
 $\|a_p\| = 1$ (unit vector)
 $Cov[PC_i, PC_p] = 0, \forall i < p$ (key point)

Assumptions

No distribution assumptions (non-parametric technique)

Big variance = most information (SNR >> 1)

PC's will be orthogonal

Solution

Using variance maximization: the searched for a_i coefficients = eigenvectors for Σ

Eigenvalue decomposition (EVD) (spectral decomposition)

Given square, symmetric, positive-definite ($p \times p$) matrix A (in statistics often Σ, S, ρ, R):

$$A = \sum_{i=1}^p \lambda_i e_i e_i^T = E \Lambda E^T$$

$$E = \begin{bmatrix} \uparrow & \dots & \uparrow \\ e_1 & \dots & e_p \\ \downarrow & \dots & \downarrow \end{bmatrix} \text{ and } E \text{ is orthogonal } (E^T = E^{-1})$$

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_p \end{bmatrix}$$

$\lambda_i, i = 1..p$ the eigenvalues are solutions to characteristic equation $|A - \lambda I| = 0$
 $e_i, i = 1..p$ the eigenvectors are solutions to eigenvalue problems $Ae_i = \lambda_i e_i, \|e_i\| = 1$

EVD for covariance matrix

$$\Sigma = E\Lambda E^T = \sum_{i=1}^p \lambda_i e_i e_i^T, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0 \quad (\text{sorted in decreasing order})$$

Terminologi and results

"loadings" = eigenvector elements

"scores" $PC_{ij} = e_i^T X_j$

$V[PC_i] = \lambda_i$

"total variance" = $\text{trace}(\Sigma) = \sum_{i=1}^p \sigma_i^2 = \sum_{i=1}^p \lambda_i$

"variance explained by PC_i " = $\frac{\lambda_i}{\sum_{i=1}^p \lambda_i} 100\%$

$\text{Cov}[PC_i, PC_k] = \delta_{ik} \Sigma$ (PC's are uncorrelated)

Geometrical interpretation

PCA is a linear change of basis: $PC = E * X$
(p x 1) = (p x p)(p x 1)

Sample based PCA procedure and dimension reduction (truncation)

Given data matrix

$$\mathbf{X} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1p} \\ X_{21} & X_{22} & \dots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{np} \end{bmatrix} = \begin{bmatrix} X_1^T \\ X_2^T \\ \vdots \\ X_n^T \end{bmatrix}$$

Estimate

$$\hat{\Sigma} = S = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^T$$

EVD

$$S = \sum_{i=1}^p \hat{\lambda}_i \hat{e}_i \hat{e}_i^T, \quad \hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_p \geq 0$$

Truncate

$$\{PC_1, PC_2, \dots, PC_p\} \rightarrow \{PC_1, PC_2, \dots, PC_r\}, \quad r < p$$

e.g so

$$\frac{\sum_{i=1}^r \hat{\lambda}_i}{\sum_{i=1}^p \hat{\lambda}_i} 100\% = 75\% - 90\% \quad (\text{other rules of thumb exist})$$

Loadings of PC_i on original variables = $\hat{e}_i$ (to help interpretation of PC's)

Scores

$$PC_{ij} = \hat{e}_i^T X_j$$

Biplot of loadings and scores

Further details

Correlation between PC's and original variables $r_{PC_i, X_k} = \hat{e}_{ik} \sqrt{\frac{\hat{\lambda}_i}{s_{kk}}}$

What does "variance explained" mean more formally/precisely (blackboard)

Applications of PCA

Data simplification / dimension reduction (lossy compression)

Interpretation of PC's from loadings and correlation with original variables

PC's can be used as predictor variables in linear regression if originals are strongly correlated

Pitfalls of PCA on S

If original variables have different units and/or very different dynamic range:

Standardize the data ("zscore", more details on blackboard) or

Do PCA on R, sample correlation matrix, instead of on S (more details on blackboard)